

1 Ecologists investigated the size of an insect population on a small island. They used a mark-release-recapture method. To mark the insects they used a fluorescent powder. This powder glows bright red when exposed to ultraviolet (UV) light.

0 1 . 1 The ecologists captured insects from a number of sites on the island. Suggest how they decided where to take their samples. [2 marks]

0 1 . 2 Give **two** assumptions made when using the mark-release-recapture method. [2 marks]

1 _____

2 _____

0 1 . 3 Suggest the advantage of using the fluorescent powder in this experiment. [2 marks]

The ecologists did **not** release any of the insects they captured 1–5 days after release of the marked insects.

Table 1 shows the ecologists' results.

Table 1

Days after release	Number of marked insects remaining in population	Number of insects captured	Number of captured insects that were marked
1	1508	524	78
2	1430	421	30
3	1400	418	18
4	1382	284	2
5	1380	232	9

0 1

4

Calculate the number of insects on this island 1 day after release of the marked insects.

Show your working.

[2 marks]

Answer = _____

0 1

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The ecologists expected to obtain the same result from their calculations of the number of insects on this island on each day during the period 1–5 days after release. In fact, their estimated number increased after day 1.

During the same period, the number of insects they caught decreased.

The method used by the ecologists might have caused these changes.

Use the information provided to suggest **one** way in which the method used by the ecologists might have caused the increase in their estimates of the size of the insect population.

[2 marks]

